InsightfulCell: Project Summary

CDU IT Code Fair Sydney 2026 · Coding Competition · Rupesh Ghimire (Product Owner and Lead Engineer) and Sandesh Prasad Paudel (Quality Assurance and Deployment Engineer), Master of Data Science, Charles Darwin University · 25 October 2026

Live application: https://app.ic.rgproject.me · **Source code:** https://github.com/rupeshghimire7/InsightfullCell · **Demo video:** [YouTube link] · **Contact:** rupeshghimire17@gmail.com

InsightfulCell is a one-stop solution for data analysis: upload a spreadsheet and receive a cleaned dataset, statistically tested findings, checked charts and maps, and a plain-language report in which every number traces back to the test that produced it.

Organisations of every size now hold more data than they have people to analyse it. They face a trust gap. General AI assistants are easy to use, but they can invent numbers, hide how they cleaned the data and need the raw rows. Business intelligence and statistics tools are accurate, but they need a trained analyst to prepare the data and choose the tests.

InsightfulCell closes that gap with a strict division of labour. Code computes every number: parsing, cleaning, statistical tests, chart values and map values. An AI model, when one is switched on, only plans the questions and phrases the sentences, and everything it writes is verified against stored facts before anyone sees it. Raw rows never reach an AI model, and the product works fully without one.

Key results

Result	Evidence
14 seconds from upload to report	5,000 rows × 19 columns on the 1 GB live server, 22 October 2026
0 of 22 cross-tenant attacks succeeded	99-check security test, 18 October 2026
31 of 31 planted errors rejected	Sentence verifier evaluation, run with every test suite
1,439 chart and map values recomputed, 0 withheld	Zomato case study, 25 October 2026
1,275 automated tests; 93.04% backend coverage	Quality gate, 24 October 2026
US$0 model cost on the live site	Rules plan the analysis and the engine writes the report

InsightfulCell serves small and medium businesses, community organisations, students, researchers and consultants who need defensible answers without an analyst. It runs on a single free-tier cloud server for about US3.60*amonth*(*estimated*), *andwithanAImodelswitchedon*, *ameasuredanalysiscostsaboutUS*0.02 in model fees.

Next, users will ask questions of their data in plain language and the service gains staging, monitoring and rate limits (Release 1 alpha, 11 December 2026), followed by team workspaces and shared reports (Release 2 beta, 12 March 2027).